A BÉZOUT DOMAIN THAT IS NOT AN ELEMENTARY DIVISOR DOMAIN

CHRISTIAN HÄGG AND ANDERS MÖRTBERG

Abstract. We settle in the negative the longstanding question whether every Bézout domain is an elementary divisor domain by constructing a Bézout domain over which an explicit 2×2 matrix has no Smith normal form. The obstruction is topological and is detected by the Möbius line bundle.

1. Introduction

The Smith normal form over a principal ideal domain is a fundamental structure theorem: every matrix can be diagonalized by invertible row and column operations, with diagonal entries forming a divisibility chain. In 1943, Helmer asked whether this conclusion continues to hold over every Bézout domain [4].[1] We answer this question in the negative.

Throughout, rings are commutative with identity, and ring homomorphisms preserve the identity. A *Bézout domain* is an integral domain in which every finitely generated ideal is principal; equivalently, every two-generated ideal is principal. For a ring A, two $m \times n$ matrices F and G over A are *equivalent* if $PFQ = G$ for some $P \in \mathrm{GL}_m(A)$ and $Q \in \mathrm{GL}_n(A)$. A *Smith normal form* of F is an equivalent diagonal matrix whose nonzero diagonal entries occur first and satisfy $d_1 \mid d_2 \mid \cdots \mid d_r$. A domain is an *elementary divisor domain* if every matrix over it has a Smith normal form. Every elementary divisor domain is Bézout, as follows by applying the definition to 1×2 matrices. The problem is whether the converse holds.

Helmer proved the elementary divisor theorem for adequate domains [4, Theorem 3]. Kaplansky subsequently introduced elementary divisor rings and showed that, over a domain, the problem is already detected by 2×2 matrices [6, Theorem 5.1 and the subsequent remark]. Gillman and Henriksen constructed a Bézout ring with zero divisors that is not an elementary divisor ring [3, Example 4.11], but the domain case remained open.

Our construction starts from

$$A_0 = \mathbb{Q}[x, y], \qquad \Delta = x^2 + y^2 - 1, \qquad M = \begin{pmatrix} 1 + x & y \\ y & 1 - x \end{pmatrix}, \tag{1.1}$$

where x and y are algebraically independent over $\mathbb{Q}$.

Theorem 1.1. *There exists a Bézout domain that is not an elementary divisor domain.*

The proof constructs an ascending sequence of smooth finitely generated factorial $\mathbb{Q}$-domains A_n together with compact sets $K_n \subset \mathrm{Spec}(A_n)(\mathbb{R})$. Every transition map $K_{n+1} \to K_n$ is a monotone surjection, so the pullback of the Möbius line bundle on the real unit circle remains nonorientable. At the same time, every two-generated ideal occurring at some stage becomes principal at a later stage. The union $R = \bigcup_n A_n$ is therefore a Bézout domain. An equivalence of M to Smith normal form over R would already be defined at one finite stage and would yield a nowhere-zero section of the corresponding nonorientable line bundle.

The algebraic point is to principalize a two-generated ideal without destroying the topological obstruction. We use the weighted invariant of Abramovich, Temkin, and Włodarczyk [1], in the presentation of Brais [2]. For a center of codimension at least two, we form its extended Rees algebra

Date: October 15, 2026.

[1] Helmer uses the term "Prüfer ring" for what we here call "Bézout domain".

and apply Jouanolou's device. The resulting smooth factorial affine ring factors the ideal by a prime power and lowers the invariant or removes one component of its maximal locus. Divisorial centers are handled by division. On real points, the torsor step admits a compact lift with connected fibers.

Outline. Section 2 gives the topological obstruction, Section 3 records the algebraic tools, Section 4 proves the principalization extension used in the construction, and Section 5 constructs the Bézout domain and proves Theorem 1.1.

AI assistance and Lean formalization. OpenAI's GPT-6 Astra (via Codex) was used in preparing this article and its Lean formalization. Anthropic's Claude Opus 5.5 was used to first (unsuccessfully) search for a counterexample and then to check the counterexample produced by GPT-6 Astra. An accompanying Lean formalization is available at `https://github.com/Zelaron/bezout-counterexample-lean`.

2. The topological obstruction

For elements $a_1, \dots, a_s$ of a ring, $(a_1, \dots, a_s)$ denotes the ideal they generate. Direct calculation from (1.1) gives

$$\det M = -\Delta, \qquad (1+x, y, 1-x) = A_0, \qquad M^2 - 2M = \Delta I_2. \tag{2.1}$$

The middle identity follows from $(1+x)+(1-x) = 2 \in \mathbb{Q}^\times$.

2.1. **Real points.** For a ring A, let $\mathrm{Spec}(A)(\mathbb{R})$ be the set of ring homomorphisms $z : A \to \mathbb{R}$, with the coarsest topology for which every evaluation map $z \mapsto z(a)$ is continuous. A homomorphism $f : A \to B$ induces the continuous map $f^* : \mathrm{Spec}(B)(\mathbb{R}) \to \mathrm{Spec}(A)(\mathbb{R})$, $z \mapsto z \circ f$. If $A = \mathbb{Q}[a_1, \dots, a_m]$ is finitely generated, then $z \mapsto (z(a_1), \dots, z(a_m))$ is a homeomorphism onto the real zero set in $\mathbb{R}^m$ of the relations among the a_i. Thus $\mathrm{Spec}(A)(\mathbb{R})$ is the usual space of real points with its Euclidean topology. In particular, it is Hausdorff, and a subset of $\mathrm{Spec}(A)(\mathbb{R})$ is compact if and only if it is closed and the generators $a_1, \dots, a_m$ are bounded on it.

A continuous surjection $q : K' \to K$ is *monotone* if every fiber $q^{-1}(z)$ is connected. In our applications, K' and K are compact Hausdorff spaces.

2.2. **The Möbius line bundle.** Let K be a topological space, and let $X, Y : K \to \mathbb{R}$ be continuous functions satisfying $X^2 + Y^2 = 1$. Write $M(X, Y)$ for the matrix obtained from M by substituting X and Y for x and y. For every $z \in K$, the matrix $\frac{1}{2}M(X(z), Y(z))$ is an idempotent of rank one by (2.1). Let $L_{X,Y}(z) \subset \mathbb{R}^2$ be its image. These fibers form a real line subbundle $L_{X,Y} \subset K \times \mathbb{R}^2$. We call $L_{X,Y}$ *orientable* if it admits a continuous nowhere-zero section, that is, a continuous map $v : K \to \mathbb{R}^2$ such that $0 \neq v(z) \in L_{X,Y}(z)$ for every $z \in K$. If $q : K' \to K$ is continuous, then the pullback of $L_{X,Y}$ along q is $L_{X \circ q, Y \circ q}$.

Let

$$K_0 = \{z \in \mathrm{Spec}(A_0)(\mathbb{R}) : z(\Delta) = 0\},$$

which is the unit circle via $z \mapsto (z(x), z(y))$, and let $L_0 = L_{x,y}$ be the corresponding line bundle. For $(x, y) = (\cos\theta, \sin\theta)$,

$$M = 2 \begin{pmatrix} \cos(\theta/2) \\ \sin(\theta/2) \end{pmatrix} \begin{pmatrix} \cos(\theta/2) & \sin(\theta/2) \end{pmatrix}. \tag{2.2}$$

Thus L_0 is the Möbius line bundle. Indeed, a continuous nowhere-zero section, pulled back to $[0, 2\pi]$, would have the form

$$f(\theta)(\cos(\theta/2), \sin(\theta/2))$$

for a continuous nowhere-zero function f. Equality at the endpoints would imply $f(2\pi) = -f(0)$, contradicting the intermediate value theorem. Hence L_0 is nonorientable.

Lemma 2.1. *Let $q : K' \to K$ be a monotone surjection from a compact space to a Hausdorff space, and let $X, Y : K \to \mathbb{R}$ be continuous with $X^2 + Y^2 = 1$. If $L_{X\circ q, Y\circ q}$ is orientable, then $L_{X,Y}$ is orientable.*

Proof. Let v be a continuous nowhere-zero section of $L_{X\circ q, Y\circ q}$, and put $u = v/\|v\|$. For $z \in K$, the line $L_{X,Y}(z)$ contains exactly two unit vectors. The continuous map u therefore takes values in a two-point set on the connected fiber $q^{-1}(z)$, so it is constant there. Hence $u = \bar{u} \circ q$ for a map $\bar{u} : K \to \mathbb{R}^2$ such that $\bar{u}(z)$ is a unit vector in $L_{X,Y}(z)$. Because K' is compact and K is Hausdorff, q is a closed map and hence a quotient map. Therefore $\bar{u}$ is continuous, and it orients $L_{X,Y}$. □

Equivalently, nonorientability pulls back along monotone surjections between compact Hausdorff spaces.

Lemma 2.2. *A composite of monotone surjections between compact Hausdorff spaces is monotone.*

Proof. We first note that if $q : K' \to K$ is monotone and $C \subset K$ is compact and connected, then $q^{-1}(C)$ is connected. Otherwise, write $q^{-1}(C) = F \amalg G$ with F and G nonempty and closed. Each fiber of q meets at most one of F and G, because it is connected. The compact sets $q(F)$ and $q(G)$ are therefore disjoint nonempty closed subsets whose union is C, a contradiction. Applying this observation to every fiber of a second monotone surjection proves the claim. □

3. Algebraic tools

A domain is *factorial* if it is a unique factorization domain (UFD). We repeatedly use finitely generated $\mathbb{Q}$-domains that are smooth over $\mathbb{Q}$ and factorial. Such rings are Noetherian.

3.1. **Kaplansky's condition.** A Bézout domain B satisfies *Kaplansky's condition* [6] if

$$\forall a, b, c \in B, \quad \gcd(a, b, c) = 1 \implies \exists p, q \in B : \ \gcd(pa, pb + qc) = 1.$$

Here $\gcd(a, b, c) = \gcd(a, \gcd(b, c))$, and gcds are normalized so that a unit gcd is 1. Equivalently, whenever $(a, b, c) = B$, there exist $p, q \in B$ with $(pa, pb + qc) = B$. This follows from $(a, b) = (\gcd(a, b))$ in a Bézout domain.

Kaplansky's condition is also equivalent to the following matrix property: every 2×2 matrix F whose entries generate B admits a row r and a column v with $rFv = 1$. The matrix property applied to $\left(\begin{smallmatrix} a & b \\ 0 & c \end{smallmatrix}\right)$ gives Kaplansky's condition. Conversely, write the first column of F as $(A, C)^{\mathsf{T}}$. Choose $d, r_0, s_0, u, v_0 \in B$ with $A = dr_0$, $C = ds_0$ and $ur_0 + v_0 s_0 = 1$: if $A = C = 0$, take $(d, r_0, s_0, u, v_0) = (0, 1, 0, 1, 0)$; otherwise let d generate (A, C) and cancel d in a Bézout identity for d. Then

$$H = \begin{pmatrix} u & v_0 \\ -s_0 & r_0 \end{pmatrix}, \qquad \det H = 1, \qquad HF = \begin{pmatrix} a & b \\ 0 & c \end{pmatrix}$$

for some $a, b, c \in B$. Invertible row operations preserve the entry ideal, so $(a, b, c) = B$. Kaplansky's condition gives p, q, s, t with $s(pa) + t(pb + qc) = 1$, hence $(p, q)HF(s, t)^{\mathsf{T}} = 1$.

Every elementary divisor domain satisfies this condition: a Smith normal form $PFQ = \operatorname{diag}(d_1, d_2)$ of such an F, with $d_1 \mid d_2$, gives $(d_1, d_2) = (d_1) = B$. Thus d_1 is a unit and gives a bilinear unit witness.

3.2. **Factoriality.** We use two standard facts. A polynomial ring over a UFD is a UFD [8, Tag 0BC1]. Nagata's criterion states that if A is a Noetherian domain and $p \in A$ is a prime element such that $A[1/p]$ is a UFD, then A is a UFD [8, Tag 0AFU]. Consequently, if A is a Noetherian UFD, then so is $A[T, T^{-1}] = A[T][1/T]$. Moreover, every prime element p of A remains prime in $A[T, T^{-1}]$, because $(A/pA)[T, T^{-1}]$ is a domain.

3.3. **Jouanolou rings.** Let B be a ring and let $c = (c_0, \dots, c_r) \in B^{r+1}$. Put

$$q_c = \sum_{i=0}^{r} c_i \sigma_i - 1 \in B[\sigma_0, \dots, \sigma_r].$$

The *Jouanolou ring* of c is

$$J_B(c) = B[\sigma_0, \dots, \sigma_r]/(q_c).$$

Let $W = \operatorname{Spec} B \setminus V(c_0, \dots, c_r)$. Over W, the row $(c_0, \dots, c_r)$ defines a surjection $\mathcal{O}_W^{\oplus(r+1)} \to \mathcal{O}_W$ with locally free kernel E_c of rank r. The morphism $\operatorname{Spec} J_B(c) \to W$ is an E_c-torsor, hence a Zariski-locally trivial affine-space bundle with affine total space. This is an explicit instance of Jouanolou's device [5, Lemme 1.5]. A real point of $J_B(c)$ is a pair (z, t) with $z \in \operatorname{Spec}(B)(\mathbb{R})$ and $t = (t_0, \dots, t_r) \in \mathbb{R}^{r+1}$ satisfying $\sum_{i=0}^{r} t_i z(c_i) = 1$.

Lemma 3.1. *Let B be a ring and $c \in B^{r+1}$.*

1. *For every homomorphism $B \to B'$, one has $J_B(c) \otimes_B B' \cong J_{B'}(c')$, where c' is the image of c. In particular, $J_B(c)/\mathfrak{a}J_B(c) \cong J_{B/\mathfrak{a}}(\bar{c})$ for every ideal $\mathfrak{a} \subset B$, and the formation of $J_B(c)$ commutes with localization of B.*
2. *For every $j \in \{0, \dots, r\}$, eliminating σ_j gives $J_B(c)[1/c_j] \cong B[1/c_j][\sigma_i \mid i \neq j]$.*
3. *If B is a domain and $c_j \neq 0$ for some j, then $J_B(c)$ is a domain and $B \to J_B(c)$ is injective.*
4. *$J_B(c)$ is formally smooth over B. If B is a smooth finitely generated $\mathbb{Q}$-algebra, then so is $J_B(c)$.*
5. *If B is a Noetherian UFD, c_j is a prime element of B, and $c_j \nmid c_m$ for some m, then $J_B(c)$ is a UFD.*

Proof. Assertion (1) follows from the right exactness of the tensor product, and (2) from solving $q_c = 0$ for σ_j.

For (3), the constant term of q_c is -1. For every $b \in B$, the image of q_c in $(B/bB)[\sigma_0, \dots, \sigma_r]$ is therefore a unit in the corresponding formal power series ring, and hence a nonzerodivisor in the polynomial ring. If B is a domain and $0 \neq b \in B$, it follows that multiplication by b is injective on $J_B(c)$: from $bf = gq_c$ in $B[\sigma_0, \dots, \sigma_r]$, reduction modulo b gives $g \in bB[\sigma_0, \dots, \sigma_r]$, and cancellation of b gives $f \in (q_c)$. Applying this to the powers of c_j shows that $J_B(c) \to J_B(c)[1/c_j]$ is injective. By (2), the target is a polynomial ring over the domain $B[1/c_j]$ and contains B.

For (4), put $P = B[\sigma_0, \dots, \sigma_r]$. Since q_c has constant term -1, it is a unit in $B[[\sigma_0, \dots, \sigma_r]]$ and hence a nonzerodivisor in P. The B-derivation $\mathcal{E} = \sum_{i=0}^{r} \sigma_i \, \partial/\partial\sigma_i$ of P satisfies $\mathcal{E}(q_c) = q_c + 1$. It therefore induces a map $\Omega_{P/B} \otimes_P J_B(c) \to J_B(c)$ that sends dq_c to 1. Thus the conormal sequence

$$0 \longrightarrow (q_c)/(q_c^2) \longrightarrow \Omega_{P/B} \otimes_P J_B(c) \longrightarrow \Omega_{J_B(c)/B} \longrightarrow 0$$

is split exact, so $J_B(c)$ is formally smooth over B by the split-conormal criterion [8, Tag 031J]. If B is smooth and finitely generated over $\mathbb{Q}$, then $J_B(c)$ is formally smooth and of finite presentation over $\mathbb{Q}$, hence smooth.

For (5), the ring $J_B(c)[1/c_j]$ is a UFD by (2) and the facts recalled above. By (1), $J_B(c)/c_j J_B(c) \cong J_{B/c_jB}(\bar{c})$. This is a domain by (3), because B/c_jB is a domain and $\bar{c}_m \neq 0$. By (3) again, c_j is nonzero in $J_B(c)$, so c_j is a prime element of the Noetherian domain $J_B(c)$. Nagata's criterion shows that $J_B(c)$ is a UFD. □

3.4. **Marked centers and the weighted invariant.** Let A be a smooth finitely generated $\mathbb{Q}$-domain and let $\mathfrak{m}$ be a maximal ideal. Then $A_{\mathfrak{m}}$ is a regular local ring of dimension $n = \dim A$, and its residue field is a finite extension of $\mathbb{Q}$. If $x = (x_1, \dots, x_n)$ is a regular system of parameters of $A_{\mathfrak{m}}$, then $dx_1, \dots, dx_n$ is a basis of $\Omega_{A_{\mathfrak{m}}/\mathbb{Q}}$, because $\mathfrak{m}/\mathfrak{m}^2 \to \Omega_{A/\mathbb{Q}} \otimes \kappa(\mathfrak{m})$ is an isomorphism in characteristic zero. Let $\partial_1, \dots, \partial_n$ be the dual derivations. We use the convention $\mathbb{N} = \{0, 1, 2, \dots\}$.

Definition 3.2. A *marked center* at $\mathfrak{m}$ is a pair $J = (x, e)$ consisting of a regular system of parameters x of $A_{\mathfrak{m}}$ and rational weights $e_1 \geq e_2 \geq \cdots \geq e_n \geq 0$. For $t \in \mathbb{Q}$, its *weighted ideal of level* t is

$$F_t(J) = \Big(x^\alpha : \ \alpha \in \mathbb{N}^n, \quad \alpha_i = 0 \text{ if } e_i = 0, \sum_{i=1}^n e_i\alpha_i \geq t\Big) A_{\mathfrak{m}}.$$

If $I \subseteq A$ is an ideal contained in $\mathfrak{m}$, then J is *admissible* for I if $IA_{\mathfrak{m}} \subseteq F_1(J)$.

In the notation of [1, 2], the marked center J is written $(x_1^{a_1}, \ldots, x_k^{a_k})$, where $a_i = 1/e_i$ and k is the number of nonzero weights; the condition denoted there by $v_J(I) \geq 1$ is precisely admissibility as defined above. We regard every weight vector as an element of $\mathbb{Q}^{\mathbb{N}}$ by appending zeros. Let $\preceq$ be the opposite of the lexicographic order:

$$e \preceq e' \quad \Longleftrightarrow \quad e' \leq_{\text{lex}} e.$$

Write $e \prec e'$ when $e \preceq e'$ and $e \neq e'$. This is the order obtained from the lexicographic order in [1, 2] after replacing the entries a_i by the weights $e_i = 1/a_i$; with this convention, worse singularities have larger invariants. For example, if $I = (y^2 - x^3) \subset \mathbb{Q}[x, y]$ and $\mathfrak{m} = (x, y)$, then the marked center with parameters (y, x) and weights $(\frac{1}{2}, \frac{1}{3})$ is admissible, and $\text{inv}_{\mathfrak{m}}(I) = (\frac{1}{2}, \frac{1}{3}, 0, \ldots)$ corresponds to the invariant $(2, 3)$ in the notation of [1].

The formulation in [2, Sections 3–4] assumes pure positive codimension. For arbitrary nonzero ideals, we use the corresponding results in [1, Sections 5–6]. All theorem numbers for [1] refer to the published version; the numbering in arXiv:1906.07106v3 is different. For a maximal ideal $\mathfrak{m} \supseteq I$, write

$$\text{ord}_{\mathfrak{m}}(I) = \max\{b \in \mathbb{N} : IA_{\mathfrak{m}} \subseteq \mathfrak{m}^b A_{\mathfrak{m}}\}.$$

Theorem 3.3. *Let A be a smooth finitely generated $\mathbb{Q}$-domain and let $I \subset A$ be a nonzero ideal.*

(1) (Invariant and maximal center; [1, Theorem 5.3.1], [2, Theorem 3.1.1].) *For every maximal ideal $\mathfrak{m} \supseteq I$ there is an admissible marked center for I at $\mathfrak{m}$ whose weight vector is lexicographically smallest among all admissible marked centers at $\mathfrak{m}$, and all admissible marked centers with this weight vector have the same weighted ideals. We call this weight vector the* invariant $\text{inv}_{\mathfrak{m}}(I)$*, and the common weighted ideals $F_t(\mathfrak{m})$ the* maximal center *of I at $\mathfrak{m}$. All weights of $\text{inv}_{\mathfrak{m}}(I)$ are at most 1, and its first weight is $1/\text{ord}_{\mathfrak{m}}(I)$* [1, Section 5.1]*. In particular, the first weight has the form $1/a$ for a positive integer a, and if $b \geq 1$ and $IA_{\mathfrak{m}} \subseteq \mathfrak{m}^b A_{\mathfrak{m}}$, then it is at most $1/b$.*

(2) (Polynomial-localization invariance; a special case of [1, Theorem 5.1.3(3)] and [2, Proposition 3.5.1].) *Let $A \to B$ be a homomorphism to a smooth finitely generated $\mathbb{Q}$-domain, let $\mathfrak{n} \supseteq IB$ be a maximal ideal of B, and put $\mathfrak{m} = \mathfrak{n} \cap A$. Suppose that, for some $r \geq 0$ and prime ideal $\mathfrak{q}$ of $A_{\mathfrak{m}}[t_1, \ldots, t_r]$, there is an A-algebra isomorphism*

$$B_{\mathfrak{n}} \cong A_{\mathfrak{m}}[t_1, \ldots, t_r]_{\mathfrak{q}}.$$

Then $\mathfrak{m}$ is maximal and $\text{inv}_{\mathfrak{n}}(IB) = \text{inv}_{\mathfrak{m}}(I)$. This is the special case of smooth invariance used below.

(3) (The maximal locus; [1, Theorems 5.1.3(2) and 6.1.1], [2, Propositions 4.1.1 and 4.1.2].) *Assume $I \neq A$. Among the invariants $\text{inv}_{\mathfrak{m}}(I)$, $\mathfrak{m} \supseteq I$ maximal, there is a largest one $\text{maxinv}(I)$ for $\preceq$. There are prime ideals $\mathfrak{p}_1, \ldots, \mathfrak{p}_r$ with $\mathfrak{p}_i + \mathfrak{p}_j = A$ for $i \neq j$ such that $\text{inv}_{\mathfrak{m}}(I) = \text{maxinv}(I)$ exactly when $\mathfrak{m}$ contains some $\mathfrak{p}_i$. If k is the number of nonzero entries of $\text{maxinv}(I)$ and $\mathfrak{m} \supseteq \mathfrak{p}_i$, there is a regular system of parameters x of $A_{\mathfrak{m}}$ with $\mathfrak{p}_i A_{\mathfrak{m}} = (x_1, \ldots, x_k)$ such that $(x, \text{maxinv}(I))$ is admissible for I at $\mathfrak{m}$; by (1), its weighted ideals are the $F_t(\mathfrak{m})$. We call $V(\mathfrak{p}_1), \ldots, V(\mathfrak{p}_r)$ the* components *of the maximal locus.*

(4) (The global center; [2, Proposition 4.1.1 and Definition 4.1.4], together with affine gluing.) *Let* $\mathfrak{p}$ *be one of the* $\mathfrak{p}_i$*. For the center obtained by retaining the component* $V(\mathfrak{p})$ *of the global associated center, put*

$$F_t = \{f \in A : \ f \in F_t(\mathfrak{m}) \text{ for every maximal ideal } \mathfrak{m} \supseteq \mathfrak{p}\} \qquad (t \in \mathbb{Q}).$$

Indeed, restrict Brais's global associated center to an open neighborhood of $V(\mathfrak{p})$ *disjoint from the other components of the maximal locus and, level by level, glue its ideals to the unit ideal on* $\operatorname{Spec}(A) \setminus V(\mathfrak{p})$*. The displayed intersection is the ideal of global sections of the resulting level-t ideal sheaf. Hence* $F_t A_{\mathfrak{m}} = F_t(\mathfrak{m})$ *for every maximal ideal* $\mathfrak{m} \supseteq \mathfrak{p}$*, and* $F_t A_{\mathfrak{m}} = A_{\mathfrak{m}}$ *for every maximal ideal* $\mathfrak{m} \not\supseteq \mathfrak{p}$*.*

(5) (Well-ordering; [1, Section 5.1], [2, Proposition 3.3.3].) *There is a set* Γ *of weight vectors, independent of* A *and* I*, that contains all invariants and such that, for every* N*, its elements with at most* N *nonzero entries are well-ordered by* $\preceq$*. This is the reciprocal, zero-padded reformulation of the sets* Γ_j *constructed in* [2, Section 3.3, especially Proposition 3.3.3].

In (3), the maximal locus is the set of closed points of $V(\mathfrak{p}_1) \amalg \cdots \amalg V(\mathfrak{p}_r)$; it is smooth of codimension k, and each $V(\mathfrak{p}_i)$ is one of its connected components.

Justification of the enlargement of Γ *in (5).* The set described in (5) may be larger than the set of actual invariants. We explain why it contains the latter and is still well-ordered in bounded length. In the differential construction of [2, Sections 3.2–3.3], after weights $e_1, \dots, e_r$ have been chosen, the next positive weight has the form

$$e_{r+1} = \frac{1 - \sum_{i=1}^{r} \beta_i e_i}{q}, \qquad \beta_i \in \mathbb{N}, \quad q \in \mathbb{N}_{>0}, \quad 0 < 1 - \sum_{i=1}^{r} \beta_i e_i \leq 1.$$

Let D be the product of the denominators, in lowest terms, of $e_1, \dots, e_r$. The numerator is h/D for an integer $1 \leq h \leq D$, and therefore

$$e_{r+1} = \frac{h}{Dq} = \frac{D!}{a}, \qquad a = \frac{D! Dq}{h} \in \mathbb{N}_{>0}.$$

For the first weight, the empty product gives $D = 1$ and $e_1 = 1/a$.

For a fixed prefix, the possible next positive weights lie in $\{D!/a : a \in \mathbb{N}_{>0}\}$, which is well-ordered by the reverse of the usual numerical order. Allowing a zero entry, after which every entry is zero, adds a largest element for this order. In a strictly descending sequence for $\preceq$, the first entries must therefore eventually stabilize; once a prefix has stabilized, the next entries must eventually stabilize as well. For sequences of length at most N, induction through the N positions rules out an infinite strictly descending sequence. This proves the required well-ordering. □

Notation 3.4. Let A be as in Theorem 3.3, let $I \subsetneq A$ be a nonzero ideal, let $V(\mathfrak{p})$ be a component of the maximal locus, and let $e = \operatorname{maxinv}(I)$ have k nonzero entries. Choose an integer $d \geq 1$ with $de_i \in \mathbb{Z}$ for all i, and index the global center of Theorem 3.3(4) by integers: $\mathcal{F}_j = F_{j/d}$ for $j \in \mathbb{Z}$. Put

$$\mathcal{R} = \bigoplus_{j \in \mathbb{Z}} \mathcal{F}_j T^j \subset A[T, T^{-1}], \qquad s = T^{-1},$$

$$\mathcal{R}_+ = (\mathcal{F}_j T^j : j \geq 1)\mathcal{R}, \qquad I_w = (fT^d : f \in I)\mathcal{R}.$$

Thus $\mathcal{R}$ is the extended Rees algebra of the weighted center along $V(\mathfrak{p})$, s is the exceptional element, and I_w is the weak transform of I. For $1 \leq j \leq d$, let G_j be a finite set of generators of $\mathcal{F}_j$.

Lemma 3.5. *In the situation of Notation 3.4:*

(1) $\mathcal{F}_j = A$ *for* $j \leq 0$*,* $\mathcal{F}_1 = \mathfrak{p}$*,* $\mathfrak{p}^j \subseteq \mathcal{F}_j$ *for* $j \geq 0$*, and* $I \subseteq \mathcal{F}_d$*. In particular,* $I_w \subseteq \mathcal{R}$*,* $I\mathcal{R} = s^d I_w$*, and* $\mathfrak{p} = \mathcal{F}_1 \subseteq s\mathcal{R}$*.*

(2) $\mathcal{F}_j = \sum_{q=1}^{d} \mathcal{F}_q \mathcal{F}_{j-q}$ *for* $j > d$. *Consequently,* $\mathcal{R} = A[s,\ gT^j :\ g \in G_j,\ 1 \le j \le d]$, *and the elements* gT^j *with* $g \in G_j$ *and* $1 \le j \le d$ *generate* $\mathcal{R}_+$.
(3) $\mathcal{R}[1/s] = A[T, T^{-1}]$.
(4) *Let* $\mathfrak{m} \supseteq \mathfrak{p}$ *be a maximal ideal, let* x *be as in Theorem 3.3(3), and put* $w_i = de_i$ *for* $i \le k$. *Then* $u_i \mapsto x_i T^{w_i}$ *induces an isomorphism*

$$A_{\mathfrak{m}}[s, u_1, \dots, u_k]/(x_i - s^{w_i} u_i \mid 1 \le i \le k) \cong \mathcal{R} \otimes_A A_{\mathfrak{m}}.$$

If $\mathfrak{m} \not\supseteq \mathfrak{p}$, *then* $\mathcal{R} \otimes_A A_{\mathfrak{m}} = A_{\mathfrak{m}}[T, T^{-1}]$. *In particular,* $\mathcal{R}$ *is a smooth finitely generated* $\mathbb{Q}$*-domain.*
(5) $\mathcal{R}/s\mathcal{R} \cong \bigoplus_{j \ge 0} \mathcal{F}_j/\mathcal{F}_{j+1}$ *is a domain. Hence* s *is a prime element of* $\mathcal{R}$.

Proof. By Theorem 3.3(4), every assertion about the ideals $\mathcal{F}_j$ may be checked after localization at maximal ideals. If $\mathfrak{m} \not\supseteq \mathfrak{p}$, then $\mathcal{F}_j A_{\mathfrak{m}} = A_{\mathfrak{m}}$ for every j. If $\mathfrak{m} \supseteq \mathfrak{p}$, then $\mathcal{F}_j A_{\mathfrak{m}} = F_{j/d}(\mathfrak{m})$ is generated by the monomials x^α with $\sum_{i=1}^{k} w_i \alpha_i \ge j$, by Theorem 3.3(3). All weights are at most 1 by Theorem 3.3(1), so the weights w_i are integers in $\{1, \dots, d\}$. This gives $\mathcal{F}_1 A_{\mathfrak{m}} = (x_1, \dots, x_k) = \mathfrak{p} A_{\mathfrak{m}}$ and $\mathfrak{p}^j \subseteq \mathcal{F}_j$, while $I \subseteq \mathcal{F}_d$ is admissibility. Since $f = s^d \cdot fT^d$ for $f \in I$ and $g = s \cdot gT$ for $g \in \mathcal{F}_1$, we have $I\mathcal{R} = s^d I_w$ and $\mathcal{F}_1 \subseteq s\mathcal{R}$. This proves (1).

For (2), a monomial of weight at least $j > d$ is the product of some x_i, which has weight $w_i \in \{1, \dots, d\}$, and a monomial of weight at least $j - w_i$. By induction on j, every homogeneous element of $\mathcal{R}$ of positive degree lies in the ideal generated by the elements gT^j and is a polynomial in them with coefficients in A. Moreover, $\mathcal{F}_j T^j = As^{-j}$ for $j \le 0$. Assertion (3) holds because $\mathcal{F}_j = A$ for $j \le 0$.

For (4), the formation of $\mathcal{R}$ commutes with localization, so $\mathcal{R} \otimes_A A_{\mathfrak{m}}$ is the extended Rees algebra of the filtration $(F_{j/d}(\mathfrak{m}))_{j \in \mathbb{Z}}$. For a filtration given by monomials in part of a regular system of parameters, the displayed presentation is [7, Proposition 5.2.2]; see also [2, Equation (2.3)]. Its left-hand side is formally smooth over $\mathbb{Q}$ by the criterion [8, Tag 031J] already used in Lemma 3.1(4): the derivations $\partial_1, \dots, \partial_k$ of $A_{\mathfrak{m}}$, extended to $A_{\mathfrak{m}}[s, u_1, \dots, u_k]$ by $\partial_i(s) = \partial_i(u_j) = 0$, satisfy $\partial_i(x_j - s^{w_j} u_j) = \delta_{ij}$. Every localization of $\mathcal{R}$ at a prime ideal is a localization of some $\mathcal{R} \otimes_A A_{\mathfrak{m}}$, so it is formally smooth over $\mathbb{Q}$. Since $\mathcal{R}$ is of finite type over $\mathbb{Q}$ by (2), it is smooth. It is a domain because it is contained in $A[T, T^{-1}]$.

For (5), the identification $\mathcal{R}/s\mathcal{R} \cong \bigoplus_j \mathcal{F}_j/\mathcal{F}_{j+1}$ follows from $s\mathcal{R} \cap \mathcal{F}_j T^j = \mathcal{F}_{j+1} T^j$. By (4), for every maximal ideal $\mathfrak{m} \supseteq \mathfrak{p}$ the graded ring $\bigoplus_j \mathcal{F}_j/\mathcal{F}_{j+1}$ becomes the polynomial ring $(A/\mathfrak{p})_{\mathfrak{m}/\mathfrak{p}}[u_1, \dots, u_k]$ after localization at $\mathfrak{m}$. In particular, each $\mathcal{F}_j/\mathcal{F}_{j+1}$ is a locally free $A/\mathfrak{p}$-module, hence torsion-free. Now let $f \in \mathcal{F}_a \setminus \mathcal{F}_{a+1}$ and $g \in \mathcal{F}_b \setminus \mathcal{F}_{b+1}$. Their images in $(\mathcal{F}_a/\mathcal{F}_{a+1})_{\mathfrak{m}}$ and $(\mathcal{F}_b/\mathcal{F}_{b+1})_{\mathfrak{m}}$ are nonzero at every maximal ideal $\mathfrak{m} \supseteq \mathfrak{p}$. Their product in $(A/\mathfrak{p})_{\mathfrak{m}/\mathfrak{p}}[u_1, \dots, u_k]$ is therefore nonzero, so $fg \notin \mathcal{F}_{a+b+1}$. Thus $\bigoplus_j \mathcal{F}_j/\mathcal{F}_{j+1}$ is a domain, and it is nonzero because $\mathcal{F}_1 = \mathfrak{p} \ne A$. □

Theorem 3.6 (Strict decrease of the invariant)**.** *In the situation of Notation 3.4, let* P *be a maximal ideal of* $\mathcal{R}$ *that contains* $I_w + s\mathcal{R}$ *but not* $\mathcal{R}_+$. *Then* $\operatorname{inv}_P(I_w) \prec \operatorname{maxinv}(I)$.

Proof. We use the affine extended-Rees argument underlying property (C) in [2, Section 2.2]. Let $\mathfrak{m} = P \cap A$. The weak Nullstellensatz makes $\mathfrak{m}$ maximal, and $\mathfrak{p} \subseteq \mathfrak{m}$ because $\mathfrak{p}\mathcal{R} \subseteq s\mathcal{R} \subseteq P$. Work near $\mathfrak{m}$ with compatible parameters as in Lemma 3.5(4). Let P_0 be the vertex point over $\mathfrak{m}$, defined by $s = u_1 = \cdots = u_k = 0$ and the remaining parameters of $A_{\mathfrak{m}}$. Equivalently,

$$P_0 = \left\{ \sum_j f_j T^j \in \mathcal{R} : f_0 \in \mathfrak{m} \right\}.$$

The differential construction of the invariant at the vertex gives $\operatorname{inv}_{P_0}(I_w) = e := \operatorname{maxinv}(I)$: the positive weights are those of the original center, now on the u_i, with s of weight zero. This follows

by transporting each step of the differential construction through $x_i = s^{w_i}u_i$; admissibility and the successive least derivative candidates are preserved. In this maximal center every homogeneous element fT^j, $f \in \mathcal{F}_j$, $j \geq 1$, belongs to the weighted ideal of level $1/d$.

Choose such an element $h = fT^j \notin P$. Spreading the vertex center and this membership to a principal neighborhood $D(G)$ of P_0, the local construction of the maximal center gives the following: at every maximal ideal $Q \in D(G)$ containing I_w, one has $\mathrm{inv}_Q(I_w) \preceq e$, and equality implies $h \in Q$. Indeed, in the equality case the spread center is the maximal center at Q by uniqueness, and its positive-level weighted ideals lie in Q.

The grading acts by automorphisms $\sigma_\lambda(T) = \lambda T$, $\lambda \in \mathbb{Q}^\times$, and preserves I_w. Since $s \in P$, all negative-degree terms of G vanish modulo P. The constant term of $\sigma_\lambda(G)$ modulo P is nonzero, because $G \notin P_0$. Thus the image of $\sigma_\lambda(G)$ is a nonzero polynomial in λ over the field $\mathcal{R}/P$, and some $\lambda \in \mathbb{Q}^\times$ makes it nonzero. Then $Q = \sigma_\lambda^{-1}(P)$ lies in $D(G)$, contains I_w, and has the same invariant as P. But $\sigma_\lambda(h) = \lambda^j h \notin P$, so $h \notin Q$. Equality with e is impossible, and $\mathrm{inv}_P(I_w) \prec e$ follows. □

Lemma 3.7. *Let $\mathfrak{m} \supseteq I$ be a maximal ideal of A, and let δ be a $\mathbb{Q}$-derivation of A with $\delta(I) \subseteq I$.*

(1) *The derivation δ preserves the maximal center: $\delta(F_t(\mathfrak{m})) \subseteq F_t(\mathfrak{m})$ for every t. Consequently, $\delta(\mathcal{F}_j) \subseteq \mathcal{F}_j$ in Notation 3.4.*

(2) *Suppose that $\delta_1, \ldots, \delta_m$ are derivations of A with $\delta_j(I) \subseteq I$, and that $\det(\delta_j(z_q))_{1\leq j,q\leq m} \notin \mathfrak{m}$ for some $z_1, \ldots, z_m \in A$. Then $\mathrm{inv}_{\mathfrak{m}}(I)$ has at most $\dim A - m$ nonzero entries.*

Proof. For (1), put $C = A_{\mathfrak{m}}$, write $e = \mathrm{inv}_{\mathfrak{m}}(I)$, and let k be the number of its nonzero entries. Choose parameters $x = (x_1, \ldots, x_n)$ compatible with the maximal center and an integer $d \geq 1$ such that every de_i is integral. Choose an integer $D > d$. Put

$$B = A[\tau], \qquad \mathfrak{n} = \mathfrak{m}B + \tau B, \qquad I_D = IB + \tau^D B.$$

Let $G_t \subseteq B_{\mathfrak{n}}$ be the weighted ideal for the ordered parameters

$$(x_1, \ldots, x_k, \tau, x_{k+1}, \ldots, x_n)$$

with weights $(e_1, \ldots, e_k, 1/D, 0, \ldots, 0)$.

Claim. The filtration (G_t) is the maximal center of I_D at $\mathfrak{n}$.

Proof of the claim. Write $a_i = 1/e_i$, so $a_i \leq d < D$, and use the differential construction of [2, Sections 3.2–3.3]. After a prefix $a_1, \ldots, a_r$ has been chosen, the next exponent is the least candidate of the form

$$\frac{\sum_{j>r} \beta_j}{1 - \sum_{i=1}^r \beta_i/a_i}, \qquad 1 - \sum_{i=1}^r \beta_i/a_i > 0,$$

for which the corresponding derivative ideal is the unit ideal; the variables include τ. For $IB_{\mathfrak{n}}$, Theorem 3.3(1)–(2) shows that the maximal center is the smooth pullback of the original center, with τ of weight zero, so its positive exponents are $a_1, \ldots, a_k$. At every stage, the derivative ideal for $I_D B_{\mathfrak{n}} = IB_{\mathfrak{n}} + \tau^D B_{\mathfrak{n}}$ is the sum of the derivative ideals for the two summands. The derivative ideal contributed by $\tau^D B_{\mathfrak{n}}$ is proper unless $\beta_\tau \geq D$, and in that case the displayed candidate is at least D. Since the sum of two proper ideals in the local ring $B_{\mathfrak{n}}$ is proper, adjoining τ^D does not alter the first k exponents, all of which are at most $d < D$. After those steps, $IB_{\mathfrak{n}}$ is admissible for the pulled-back center, while τ^D contributes the next exponent D and the parameter τ, obtained by differentiating $D - 1$ times. The enlarged center is admissible for $I_D B_{\mathfrak{n}}$, so the construction stops. Uniqueness of the maximal center identifies its filtration with (G_t), proving the claim.

Extend δ to C and set $\delta(\tau) = 0$. Under the identification

$$B_{\mathfrak{n}}/\tau^D B_{\mathfrak{n}} \cong C[\tau]/(\tau^D),$$

the formulas

$$\Phi(\tau) = \tau, \qquad \Phi(f) = \sum_{j=0}^{D-1} \frac{\tau^j \delta^j(f)}{j!} \quad (f \in C)$$

define an automorphism; its inverse is obtained by replacing τ by $-\tau$. Since $\delta(IC) \subseteq IC$, the automorphism Φ preserves the image of I_D. Put

$$x_i' = \sum_{j=0}^{D-1} \frac{\tau^j \delta^j(x_i)}{j!}.$$

Modulo $(\mathfrak{n}B_\mathfrak{n})^2$, each x_i' differs from x_i by a scalar multiple of τ. Hence

$$(x_1', \ldots, x_k', \tau, x_{k+1}', \ldots, x_n')$$

is a regular system of parameters. Let G_t' be the weighted ideals defined by these parameters and the same weights as G_t. Modulo τ^D, the automorphism Φ preserves the image of I_D and carries the image of G_1 to that of G_1'. Therefore

$$I_D B_\mathfrak{n} \subseteq G_1' + \tau^D B_\mathfrak{n} = G_1',$$

where the equality follows from $\tau^D \in G_1'$. Thus the new center is admissible for I_D. The claim identifies its weight vector as maximal, so uniqueness gives $G_t' = G_t$ for every t.

In particular, $x_i' \in G_{e_i}$ for $1 \le i \le k$. Under the natural map $B_\mathfrak{n} \to C[[\tau]]$, expand an element of G_{e_i} in the monomial generators of that ideal. Only generators of τ-degree zero or one can contribute to the coefficient of τ. In the first case the x-weight is at least e_i, and in the second it is at least $e_i - 1/D$. Hence the coefficient belongs to $F_{e_i - 1/D}(\mathfrak{m})$, and therefore

$$\delta(x_i) \in F_{e_i - 1/D}(\mathfrak{m}) = F_{e_i}(\mathfrak{m}).$$

The equality holds because e_i and every x-monomial weight lie in $\frac{1}{d}\mathbb{Z}$, whereas $1/D < 1/d$. The Leibniz rule now gives $\delta(F_t(\mathfrak{m})) \subseteq F_t(\mathfrak{m})$ for every t. Applying this at every maximal ideal containing the chosen component and using the definition in Theorem 3.3(4) proves $\delta(\mathcal{F}_j) \subseteq \mathcal{F}_j$.

For (2), let k be the number of nonzero entries of $e = \mathrm{inv}_\mathfrak{m}(I)$. For $0 < \varepsilon \le e_k$ we have $F_\varepsilon(\mathfrak{m}) = (x_1, \ldots, x_k)$, so by (1) every $\delta_j(x_i)$ with $i \le k$ lies in $\mathfrak{m}$. Writing $\delta_j = \sum_i \delta_j(x_i)\partial_i$, we see that the matrix $(\delta_j(z_l))$ is the product of $(\delta_j(x_i))_{j,i}$ and $(\partial_i z_l)_{i,l}$. Modulo $\mathfrak{m}$, the first factor has zero columns for $i \le k$, so the product has rank at most $n - k$ over $\kappa(\mathfrak{m})$. Hence $m \le n - k$. □

4. Principalizing ideals while preserving the obstruction

Throughout this section, A is a smooth finitely generated factorial $\mathbb{Q}$-domain, and $I \subsetneq A$ is a nonzero ideal. We fix a component $V(\mathfrak{p})$ of the maximal locus of I and use Notation 3.4; in particular, $k \ge 1$ is the number of nonzero entries of $e = \mathrm{maxinv}(I)$. Since A is a UFD and $\mathfrak{p} \ne 0$, the prime ideal $\mathfrak{p}$ contains a prime element π. Let $c(I)$ denote the number of components of the maximal locus.

One step replaces (A, I) by a pair (B, I_1), where $B = A$ or B is a smooth finitely generated factorial $\mathbb{Q}$-domain containing A, and factors IB as $IB = \alpha I_1$ for some $\alpha \in B$. The complexity decreases in the following sense: either $I_1 = B$, or $\mathrm{maxinv}(I_1) \prec \mathrm{maxinv}(I)$, or $\mathrm{maxinv}(I_1) = \mathrm{maxinv}(I)$ and $c(I_1) < c(I)$. We distinguish the divisorial case $k = 1$ from the case $k \ge 2$.

4.1. **The divisorial step.**

Lemma 4.1. *Assume $k = 1$, so $e = (1/a, 0, 0, \ldots)$ for a positive integer a. Then $\mathfrak{p} = \pi A$, $I \subseteq \pi^a A$, and for $I_1 = (I : \pi^a)$ one has $I = \pi^a I_1$. Moreover, either $I_1 = A$, or $\mathrm{maxinv}(I_1) \prec \mathrm{maxinv}(I)$, or $\mathrm{maxinv}(I_1) = \mathrm{maxinv}(I)$ and $c(I_1) < c(I)$.*

Proof. At every maximal ideal containing $\mathfrak{p}$, the ideal $\mathfrak{p}$ is generated by one parameter. Thus $\mathfrak{p}$ is a height-one prime, hence $\mathfrak{p} = \pi A$ because A is factorial. Admissibility gives $I \subseteq \pi^a A$, and therefore $I = \pi^a I_1$.

No maximal ideal can contain both I_1 and π: otherwise $IA_{\mathfrak{m}} \subseteq \mathfrak{m}^{a+1}A_{\mathfrak{m}}$, so the first weight of $\mathrm{inv}_{\mathfrak{m}}(I)$ would be at most $1/(a+1)$, contradicting $\mathrm{inv}_{\mathfrak{m}}(I) = e$. Hence π is a unit at every maximal ideal containing I_1, where I_1 and I have the same localization and the same invariant. Thus the chosen component $V(\mathfrak{p})$ disappears and no new maximal component is created. The conclusion follows from Theorem 3.3(3). □

4.2. **The torsor step.** Assume $k \geq 2$. Let

$$\mathbf{h} = (h_0, h_1, \ldots, h_\ell) = (\pi T,\ gT^j :\ g \in G_j,\ 1 \leq j \leq d)$$

be the tuple of elements of $\mathcal{R}_+$ obtained by listing πT followed by the elements gT^j with $g \in G_j$ and $1 \leq j \leq d$. Write $h_i = g_i T^{j_i}$, where $1 \leq j_i \leq d$ and $g_i \in \mathcal{F}_{j_i}$; thus $g_0 = \pi$ and $j_0 = 1$. By Lemma 3.5(2), the entries of $\mathbf{h}$ generate $\mathcal{R}_+$. Put

$$U = J_{\mathcal{R}}(\mathbf{h}) = \mathcal{R}[\sigma_0, \ldots, \sigma_\ell]/\Big(\sum_{i=0}^{\ell} h_i\sigma_i - 1\Big),$$

so that $\operatorname{Spec} U \to \operatorname{Spec} \mathcal{R} \setminus V(\mathcal{R}_+)$ is the Jouanolou affine-space bundle associated with $\mathbf{h}$, and let $I_1 = I_w U$.

Lemma 4.2. *The ring U is a smooth finitely generated factorial $\mathbb{Q}$-domain, the homomorphism $A \to U$ is injective, and $IU = s^d I_1$.*

Proof. By Lemmas 3.5 and 3.1, the ring U is a smooth finitely generated $\mathbb{Q}$-domain, $\mathcal{R} \to U$ is injective, and $IU = s^d I_1$. Modulo s one has

$$U/sU \cong J_{\mathcal{R}/s\mathcal{R}}(\overline{\mathbf{h}}).$$

The base ring is a domain by Lemma 3.5(5), and the row $\overline{\mathbf{h}}$ is nonzero: after localizing at a maximal ideal over $\mathfrak{p}$ and writing $w_1 = de_1$, one has $x_1 \in \mathcal{F}_{w_1}A_{\mathfrak{m}} \setminus \mathcal{F}_{w_1+1}A_{\mathfrak{m}}$, so some generator in G_{w_1} has nonzero initial form. Lemma 3.1(3) therefore shows that U/sU is a domain, so s is prime in U.

After inverting s,

$$U[1/s] \cong J_{A[T,T^{-1}]}(\mathbf{h}).$$

The Laurent polynomial ring is factorial, the row contains the prime element πT, and it also contains an element gT not divisible by πT: otherwise the height-k prime $\mathfrak{p}$, with $k \geq 2$, would be generated by π. Hence Lemma 3.1(5) makes $U[1/s]$ factorial, and Nagata's criterion makes U factorial. The inclusion $A \hookrightarrow U$ follows from $A \subseteq \mathcal{R} \hookrightarrow U$. □

Lemma 4.3. *The ideal I_1 is nonzero. Moreover, either $I_1 = U$, or $\mathrm{maxinv}(I_1) \prec \mathrm{maxinv}(I)$, or $\mathrm{maxinv}(I_1) = \mathrm{maxinv}(I)$ and $c(I_1) < c(I)$.*

Proof. The ideal I_1 contains $fT^d \neq 0$ for $0 \neq f \in I$, since $\mathcal{R} \to U$ is injective. Let $Q \supseteq I_1$ be maximal and put $P = Q \cap \mathcal{R}$. The contractions P and $Q \cap A$ are maximal by the weak Nullstellensatz. Since the h_i generate the unit ideal in U, some $h_i \notin P$; hence U_Q is a localization of a polynomial ring over $\mathcal{R}_P$, and

$$\mathrm{inv}_Q(I_1) = \mathrm{inv}_P(I_w).$$

If $s \in Q$, Theorem 3.6 gives a strict decrease. If $s \notin Q$, then $\mathcal{R}_P$ is a localization of $A_{Q\cap A}[T, T^{-1}]$ by Lemma 3.5(3), so Theorem 3.3(2) gives

$$\mathrm{inv}_Q(I_1) = \mathrm{inv}_{Q\cap A}(I).$$

Moreover $Q \cap A$ cannot contain $\mathfrak{p}$, for then $s^{j_i}h_i = g_i \in \mathfrak{p} \subseteq P$ would imply $h_i \in P$ for every i.

It follows that every invariant of I_1 is at most $\operatorname{maxinv}(I)$, and equality can occur only over a component $V(\mathfrak{p}')$ of the old maximal locus with $\mathfrak{p}' \neq \mathfrak{p}$. Conversely, let Q be any maximal ideal containing $\mathfrak{p}'U$. Then $s \notin Q$: otherwise $\mathfrak{p}U \subseteq sU \subseteq Q$, contradicting $(\mathfrak{p}+\mathfrak{p}')U = U$. Since $I \subseteq \mathfrak{p}'$, the identity $IU = s^d I_1$ gives $I_1 U_Q = IU_Q \subseteq QU_Q$, so $I_1 \subseteq Q$ and the invariant at Q equals $\operatorname{maxinv}(I)$. Finally,

$$\mathcal{R}/\mathfrak{p}'\mathcal{R} \cong (A/\mathfrak{p}')[T, T^{-1}],$$

and some generator from G_1 survives modulo $\mathfrak{p}'$, so Lemma 3.1(3) shows that $U/\mathfrak{p}'U$ is a nonzero domain. Thus the ideals $\mathfrak{p}'U$, for the old components other than $\mathfrak{p}$, are proper pairwise-comaximal primes and are exactly the components that can remain. The claimed decrease follows. □

4.3. **Termination.** The torsor step raises dimension, so we keep track of the new smooth directions. Fix $N \geq 0$. Say that a triple $(B, \mathfrak{b}, m)$ satisfies $(\star_N)$ if

$$\dim B \leq N + m$$

and, at every maximal ideal $\mathfrak{m} \supseteq \mathfrak{b}$, there are m derivations preserving $\mathfrak{b}$ whose values on m elements have determinant outside $\mathfrak{m}$. By Lemma 3.7(2), every invariant of such an ideal has at most N nonzero entries. The initial triple $(A, I, 0)$ satisfies $(\star_{\dim A})$.

Lemma 4.4. *Suppose that (A, I, m) satisfies $(\star_N)$. In the divisorial step, (A, I_1, m) satisfies $(\star_N)$. In the torsor step, $(U, I_1, m + \ell + 1)$ satisfies $(\star_N)$.*

Proof. In the divisorial step, the dimension bound is unchanged and $V(I_1) \subseteq V(I)$. By Lemma 3.7(1), a derivation preserving I satisfies $\delta(\pi) = b\pi$ and therefore preserves $(I : \pi^a)$, since

$$\pi^a \delta(f) = \delta(\pi^a f) - ab\,\pi^a f \in I \qquad (\pi^a f \in I).$$

In the torsor step, $\dim U = \dim A + 1 + \ell \leq N + m + 1 + \ell$. Let $Q \supseteq I_1$ be maximal. Then $Q \cap A$ is maximal and contains I, so choose the m derivations and elements supplied by $(\star_N)$ there, and choose i_0 with $h_{i_0} \notin Q$. Lemma 3.7(1) lets each old derivation act coefficientwise on $\mathcal{R}$ and extend to U by

$$\delta(\sigma_i) = -\sigma_i \sum_r \sigma_r \delta(h_r).$$

Together with the Euler derivation

$$E(fT^q) = qfT^q, \qquad E(\sigma_i) = -j_i \sigma_i$$

and the ℓ vertical derivations

$$h_r \frac{\partial}{\partial \sigma_{i_0}} - h_{i_0} \frac{\partial}{\partial \sigma_r} \qquad (r \neq i_0),$$

these give $m + \ell + 1$ derivations preserving I_1. On the elements $z_1, \ldots, z_m, h_{i_0}$ and σ_r for $r \neq i_0$, their value matrix is block upper triangular with determinant

$$\det(\delta_j(z_q))\, j_{i_0} h_{i_0} (-h_{i_0})^{\ell} \notin Q.$$

Thus $(U, I_1, m + \ell + 1)$ satisfies $(\star_N)$. □

4.4. **Real points.**

Lemma 4.5. *In the torsor step, let $K \subset \operatorname{Spec}(A)(\mathbb{R})$ be compact, put $E_i = 2\,d!/j_i$ (an even integer), and let $K' \subset \operatorname{Spec}(U)(\mathbb{R})$ be the set of real points z' such that*

$$z'|_A \in K, \qquad z'(s) \geq 0, \qquad \sum_{i=0}^{\ell} z'(h_i)^{E_i} = 1,$$

$$z'(\sigma_i) = z'(h_i)^{E_i - 1} \qquad (0 \leq i \leq \ell).$$

Then K' is compact, and $K' \to K$ is a monotone surjection.

Proof. The last two conditions are compatible with $\sum_i h_i\sigma_i = 1$. If $z' \in K'$ and $z = z'|_A$, then $s^{j_i}h_i = g_i$ and $j_iE_i = 2d!$ give

$$z'(s)^{2d!} = \sum_{i=0}^{\ell} z(g_i)^{E_i} =: \nu(z). \tag{4.1}$$

The defining conditions are closed, $|z'(h_i)|, |z'(\sigma_i)| \le 1$, and (4.1) bounds $z'(s)$ on K'. Since U is generated over A by s, the h_i, and the σ_i, while elements of A are bounded on the compact set K, the criterion of Subsection 2.1 shows that K' is compact. The restriction map $K' \to K$ is continuous.

The function $\nu(z)$ is positive exactly when $\mathfrak{p} \not\subseteq \ker z$: every g_i lies in $\mathfrak{p}$, while the coefficients with $j_i = 1$ generate $\mathfrak{p}$. If $\nu(z) > 0$, equation (4.1) forces $z'(s) = \rho := \nu(z)^{1/(2d!)} > 0$. There is then a unique point over z: extend z to $A[T, T^{-1}]$ by $z'(T) = \rho^{-1}$ and put $z'(\sigma_i) = z'(h_i)^{E_i - 1}$.

Now suppose $\mathfrak{p} \subseteq \ker z$. Then every point over z sends s to zero. Choose a maximal ideal $\mathfrak{m} \supseteq \ker z$ and use the local presentation in Lemma 3.5(4). Points of $\mathcal{R}$ over z with $s = 0$ are parametrized by $a = (a_1, \dots, a_k) \in \mathbb{R}^k$, where a_r is the value of $x_rT^{w_r}$. Write $H_i(a)$ for the value of h_i at this point. Each H_i is weighted homogeneous of degree j_i, so, with

$$\lambda \star a = (\lambda^{w_1}a_1, \dots, \lambda^{w_k}a_k), \qquad \varphi(a) = \sum_i H_i(a)^{E_i},$$

one has $\varphi(\lambda \star a) = \lambda^{2d!}\varphi(a)$. Moreover, if $\varphi(a) = 0$, then the even exponents force every $H_i(a)$ to vanish. The corresponding point therefore vanishes on the ideal $\mathcal{R}_+$ generated by the h_i, including every $x_rT^{w_r}$, and hence $a = 0$.

The fiber of $K' \to K$ over z is therefore the level set $\varphi(a) = 1$, with the values of the σ_i determined by the defining equations. The normalization

$$a \longmapsto \varphi(a)^{-1/(2d!)} \star a$$

is a continuous surjection from $\mathbb{R}^k \setminus \{0\}$ onto this fiber. Since $k \ge 2$, the source is connected. Thus every fiber is nonempty and connected, and $K' \to K$ is a monotone surjection. □

4.5. **The principalization result.**

Proposition 4.6 (Principalization extension)**.** *Let A be a smooth finitely generated factorial $\mathbb{Q}$-domain, let $a, b \in A$ be not both zero, and let $K \subset \operatorname{Spec}(A)(\mathbb{R})$ be compact. There are a smooth finitely generated factorial $\mathbb{Q}$-domain A', an injection $A \hookrightarrow A'$, and a compact set $K' \subset \operatorname{Spec}(A')(\mathbb{R})$ such that $(a, b)A'$ is principal and the induced map $K' \to K$ is a monotone surjection.*

Proof. Put $I = (a, b)$ and $N = \dim A$. We prove the following auxiliary assertion. Let B be a smooth finitely generated factorial $\mathbb{Q}$-domain, let $\mathfrak{b} \subseteq B$ be a nonzero ideal, let $m \ge 0$ be such that $(B, \mathfrak{b}, m)$ satisfies $(\star_N)$, and let $K \subseteq \operatorname{Spec}(B)(\mathbb{R})$ be compact. Then there are an injection $B \hookrightarrow B'$ into a smooth finitely generated factorial $\mathbb{Q}$-domain and a compact set $K' \subseteq \operatorname{Spec}(B')(\mathbb{R})$ such that $\mathfrak{b}B'$ is principal and $K' \to K$ is a monotone surjection. The proposition is the initial case $(B, \mathfrak{b}, m) = (A, I, 0)$.

If $\mathfrak{b} = B$, take $B' = B$ and $K' = K$. For $\mathfrak{b} \neq B$ we argue by well-founded induction on the pair $(\operatorname{maxinv}(\mathfrak{b}), c(\mathfrak{b}))$, ordered lexicographically, with $\preceq$ on the first entry and the usual order on $\mathbb{N}$ on the second. This is well-founded by Theorem 3.3(5), because all invariants of ideals in triples satisfying $(\star_N)$ have at most N nonzero entries. Let k be the number of nonzero entries of $\operatorname{maxinv}(\mathfrak{b})$, choose a component of the maximal locus of $\mathfrak{b}$, and apply the divisorial step if $k = 1$ or the torsor step if $k \ge 2$. This gives a smooth finitely generated factorial $\mathbb{Q}$-domain B_1 with an injection $B \hookrightarrow B_1$ (namely $B_1 = B$, or $B_1 = U$ by Lemma 4.2), a compact set $K_1 \subset \operatorname{Spec}(B_1)(\mathbb{R})$ with $K_1 \to K$ a monotone surjection (with $K_1 = K$ in the divisorial case and K_1 as in Lemma 4.5 in the torsor case), an ideal $\mathfrak{b}_1 \subseteq B_1$, and an element $\alpha \in B_1$ with $\mathfrak{b}B_1 = \alpha\mathfrak{b}_1$ (where $\alpha = \pi^a$ or

$\alpha = s^d$). Let $m_1 = m$ in the divisorial step and $m_1 = m + \ell + 1$ in the torsor step. By Lemmas 4.1, 4.3 and 4.4, the triple $(B_1, \mathfrak{b}_1, m_1)$ satisfies $(\star_N)$, and either $\mathfrak{b}_1 = B_1$ or $(\mathrm{maxinv}(\mathfrak{b}_1), c(\mathfrak{b}_1))$ is strictly smaller than $(\mathrm{maxinv}(\mathfrak{b}), c(\mathfrak{b}))$ in this lexicographic order.

In either case, the assertion for $(B_1, \mathfrak{b}_1, m_1)$ and K_1 holds, trivially if $\mathfrak{b}_1 = B_1$ and by induction otherwise. It gives $B_1 \hookrightarrow B'$ and a compact set $K' \subset \mathrm{Spec}(B')(\mathbb{R})$ with $\mathfrak{b}_1 B' = \alpha' B'$ and $K' \to K_1$ a monotone surjection. Then $\mathfrak{b} B' = \alpha\alpha' B'$ is principal, $B \to B'$ is injective, and $K' \to K$ is a monotone surjection by Lemma 2.2. □

5. Construction of the Bézout domain

Construction 5.1. Fix a bijection $\langle \cdot, \cdot \rangle : \mathbb{N} \times \mathbb{N} \to \mathbb{N}$ with $i \le \langle i, j \rangle$ for all i, j, for instance the Cantor pairing. Starting with $A_0 = \mathbb{Q}[x, y]$ and the circle K_0, we construct injections

$$A_0 \hookrightarrow A_1 \hookrightarrow A_2 \hookrightarrow \cdots$$

of smooth finitely generated factorial $\mathbb{Q}$-domains and compact sets $K_n \subset \mathrm{Spec}(A_n)(\mathbb{R})$ such that each induced map $K_{n+1} \to K_n$ is a monotone surjection.

Every finitely generated $\mathbb{Q}$-algebra is countable. Once A_i has been constructed, choose a surjection

$$\eta_i : \mathbb{N} \longrightarrow A_i \times A_i.$$

To pass from A_n to A_{n+1}, write $n = \langle i, j \rangle$. Then $i \le n$, and we let (a, b) be the image of the pair $\eta_i(j)$ in A_n. If $a = 0$ or $b = 0$, the ideal $(a, b)A_n$ is principal, and we put $A_{n+1} = A_n$ and $K_{n+1} = K_n$. Otherwise, write $a = ca'$ and $b = cb'$ with a' and b' coprime, which is possible because A_n is a UFD. Apply Proposition 4.6 to a', b', and the compact set K_n, and let A_{n+1} and K_{n+1} be the resulting ring and compact set. Then $(a, b)A_{n+1} = c\,(a', b')A_{n+1}$ is principal.

Identify every A_n with its image in later rings and set

$$R = \bigcup_{n \ge 0} A_n. \tag{5.1}$$

Proposition 5.2. *The ring R is a Bézout domain.*

Proof. Since all transition maps are injective, the increasing union R is a domain. Let $a, b \in R$. Choose i such that $a, b \in A_i$, and choose $j \in \mathbb{N}$ with $\eta_i(j) = (a, b)$. This pair is processed while passing from A_n to A_{n+1}, where $n = \langle i, j \rangle$. Hence

$$(a, b)A_{n+1} = gA_{n+1}$$

for some $g \in A_{n+1}$. Extending to R gives $(a, b)R = gR$. Thus R is a Bézout domain. □

The functions x and y on K_n are pulled back from K_0, so the line bundle

$$L_n = L_{x,y} \quad \text{on } K_n$$

is the pullback of L_0 along the composite of the transition maps. By Lemmas 2.1 and 2.2,

$$L_n \text{ is nonorientable for every } n \ge 0. \tag{5.2}$$

Lemma 5.3. *There are no elements $p, q, u, v, \mu \in R$ such that*

$$\mu\,(p, q)\,M \begin{pmatrix} u \\ v \end{pmatrix} = 1.$$

Proof. Suppose such elements exist, and choose N such that all five lie in A_N. The identity already holds in A_N, because $A_N \hookrightarrow R$ is injective. For $z \in K_N$, put

$$\xi(z) = M(z) \begin{pmatrix} u(z) \\ v(z) \end{pmatrix} \in L_N(z).$$

The assumed identity gives $\mu(z)(p(z), q(z))\,\xi(z) = 1$, so $\xi(z) \neq 0$. Thus ξ is a continuous nowhere-zero section of L_N, contradicting (5.2). □

Failure of Kaplansky's condition. The same ring R fails Kaplansky's condition in the gcd formulation:

$$\exists a, b, c \in R: \quad \gcd(a, b, c) = 1 \quad \text{and} \quad \forall p, q \in R: \ \gcd(pa, pb + qc) \neq 1.$$

Indeed, R is Bézout by Proposition 5.2, and the entries of M generate R by (2.1). If R satisfied Kaplansky's condition, the equivalent matrix property in Subsection 3.1 would give a bilinear unit witness for M, contradicting Lemma 5.3 with $\mu = 1$.

Proof of Theorem 1.1. Take the ring R from Construction 5.1. By Proposition 5.2, it is a Bézout domain. It remains to show that it is not an elementary divisor domain.

Suppose, to the contrary, that the matrix M has a Smith normal form over R:

$$PMQ = \operatorname{diag}(d_1, d_2), \qquad P, Q \in \mathrm{GL}_2(R), \qquad d_1 \mid d_2. \tag{5.3}$$

Multiplication by invertible matrices preserves the ideal generated by all matrix entries. By (2.1), the entries of M generate R. Hence $(d_1, d_2) = R$. Since $d_1 \mid d_2$, this ideal equals (d_1), so d_1 is a unit.

Let (p, q) be the first row of P and $(u, v)^{\mathsf{T}}$ the first column of Q. Comparing upper-left entries in (5.3) gives

$$(p, q)M(u, v)^{\mathsf{T}} = d_1.$$

Multiplying by d_1^{-1} contradicts Lemma 5.3. Thus M has no Smith normal form over R, so R is not an elementary divisor domain. □

References

[1] D. Abramovich, M. Temkin, and J. Włodarczyk, *Functorial embedded resolution via weighted blowings up*, Algebra Number Theory **18** (2024), no. 8, 1557–1587. doi:10.2140/ant.2024.18.1557.

[2] M. J.-L. Brais, *Streamlining resolution of singularities with weighted blow-ups*, preprint, arXiv:2512.01859v1 (2025).

[3] L. Gillman and M. Henriksen, *Rings of continuous functions in which every finitely generated ideal is principal*, Trans. Amer. Math. Soc. **82** (1956), 366–391. doi:10.1090/S0002-9947-1956-0078980-4.

[4] O. Helmer, *The elementary divisor theorem for certain rings without chain condition*, Bull. Amer. Math. Soc. **49** (1943), no. 4, 225–236. doi:10.1090/S0002-9904-1943-07886-X.

[5] J.-P. Jouanolou, *Une suite exacte de Mayer–Vietoris en K-théorie algébrique*, in H. Bass (ed.), *Algebraic K-Theory I: Higher K-Theories*, Lecture Notes in Mathematics, vol. 341, Springer-Verlag, Berlin–Heidelberg, 1973, pp. 293–316. doi:10.1007/BFb0067063.

[6] I. Kaplansky, *Elementary divisors and modules*, Trans. Amer. Math. Soc. **66** (1949), 464–491. doi:10.1090/S0002-9947-1949-0031470-3.

[7] M. H. Quek and D. Rydh, *Weighted blow-ups*, preprint, March 29, 2022, `https://people.kth.se/~dary/weighted-blowups20220329.pdf`.

[8] The Stacks Project Authors, *The Stacks Project*, 2026. `https://stacks.math.columbia.edu`.

Department of Mathematics, Stockholm University
Email address: `hagg@math.su.se`

Department of Mathematics, Stockholm University
Email address: `anders.mortberg@math.su.se`